

Teacher Guide — senior / module-06- moon-phases-tithi

IKS Curriculum

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Overview

Moon Phases and Tithi

Band: Senior (Grades 9–12) · **Sessions:** 10 × 45 min · **Sanskrit:** Candra-pakṣa, Tithi

Overview

The Moon’s apparent change of shape across ~29.5 days is the synodic month. Indian astronomy divides this into 30 *tithis* (lunar days), grouped into a waxing fortnight (*śukla pakṣa*) and waning fortnight (*kṛṣṇa pakṣa*). This module connects observation, lunar geometry, calendar construction, and the panchanga framework.

NEP 2020 alignment

- §4.6 — Integration of Indian Knowledge Systems
- §4.23 — Experiential learning (moon-watching)
- §11.8 — Holistic, multidisciplinary (astronomy + calendar + culture)

Key terms (this module)

- *candra* — Moon
- *pakṣa* — fortnight (half-month)
- *śukla pakṣa* — waxing fortnight (new → full)
- *kṛṣṇa pakṣa* — waning fortnight (full → new)
- *tithi* — lunar day (1/30 of synodic month)
- *amāvasyā* — new moon
- *pūrṇimā* — full moon
- *pañcāṅga* — five-limbed Indian calendar (tithi, vāra, nakṣatra, yoga, karaṇa)

Prerequisites

Module 2 (Panchabhūta) is recommended before this module. Modules 5 (Bindu Paddhati / dot addition) and 7 (×11 Multiplication) are useful prerequisites for Module 9 and 11.

Materials needed (across 10 days)

- Notebook + pencils per student
- Whiteboard / chart paper
- (See `lesson-plan.md` for per-day material lists)

10-day map

- **Day 1** — The Moon and Its Shapes: Observe and name the four basic phases (new, first quarter, full, last quarter).
- **Day 2** — Two Fortnights — Pakṣa: Distinguish śukla pakṣa from kṛṣṇa pakṣa and name today's.
- **Day 3** — Fifteen Tithis per Pakṣa: Recite the 15 tithi names (pratipadā, dvitīyā, ...) and explain why a tithi isn't exactly a day.
- **Day 4** — Building a Tithi Calendar: Construct a tithi calendar for the current month using observation and computation.
- **Day 5** — Lunar Months and Festivals: Map the 12 lunar months (caitra, vaiśākha, ...) to major festivals.
- **Day 6** — Panchanga — Five Limbs: Identify the five components: tithi, vāra, nakṣatra, yoga, karaṇa.
- **Day 7** — Why Eclipses Happen: Explain solar/lunar eclipses using orbital plane geometry (5° tilt of Moon's orbit).
- **Day 8** — Modern Lunar Science: Phases as Sun's illumination of Moon; mention libration, tides.
- **Day 9** — Cross-cultural Calendars: Compare with Hijri (Islamic), Chinese, and Hebrew lunar calendars.
- **Day 10** — Synthesis and Quiz: Final quiz and 10-day moon-diary share.

Files in this pack

- `lesson-plan.md` — detailed 10-day arc
- `sources.md` — sources cited in this module
- `teacher-notes.md` — common misconceptions, differentiation
- `parent-guide.md` — what parents should know
- `days/day-NN-*.md` — 10 per-day lesson plans
- `quizzes/formative.md`, `quizzes/summative.md`
- `activities/activity-01-*.md`
- `assessments/rubric.md`, `assessments/project-brief.md`
- `student-workbook.md`
- `writing-prompts.md` (or `oral-recall.md` for Primary)
- `homework.md`
- `slides/deck.md`

Voice

Analytical, comparative, primary-source aware; 20–30 words on average. Pitches the concept first, frames the tradition as historical context, and uses IAST for all Sanskrit terms.

10-Day Lesson Plan

Lesson Plan — Moon Phases and Tithi (Senior)

10-day arc, 45 min per session.

Day 1 — The Moon and Its Shapes

Objective: Observe and name the four basic phases (new, first quarter, full, last quarter).

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 2 — Two Fortnights — Pakṣa

Objective: Distinguish śukla pakṣa from kṛṣṇa pakṣa and name today's.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 3 — Fifteen Tithis per Pakṣa

Objective: Recite the 15 tithi names (pratipadā, dvitīyā, ...) and explain why a tithi isn't exactly a day.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 4 — Building a Tithi Calendar

Objective: Construct a tithi calendar for the current month using observation and computation.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 5 — Lunar Months and Festivals

Objective: Map the 12 lunar months (caitra, vaiśākha, ...) to major festivals.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 6 — Panchanga — Five Limbs

Objective: Identify the five components: tithi, vāra, nakṣatra, yoga, karaṇa.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 7 — Why Eclipses Happen

Objective: Explain solar/lunar eclipses using orbital plane geometry (5° tilt of Moon's orbit).

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 8 — Modern Lunar Science

Objective: Phases as Sun’s illumination of Moon; mention libration, tides.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 9 — Cross-cultural Calendars

Objective: Compare with Hijri (Islamic), Chinese, and Hebrew lunar calendars.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 10 — Synthesis and Quiz

Objective: Final quiz and 10-day moon-diary share.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Pacing notes

- Senior sessions are 45 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see [assessments/project-brief.md](#)) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see [teacher-notes.md](#) for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Teacher Notes

Teacher Notes — Moon Phases and Tithi (Senior)

Common student misconceptions

- *“This is religious, not scientific.”* — Frame the tradition as historical observation. Pañcabhūta-like classifications of matter exist in many cultures. Indian astronomy and math are documented, peer-reviewed historical fields.
- *“The Sanskrit terms are just jargon.”* — Insist on the triple format (Sanskrit / IAST / meaning) once, then use the Sanskrit alone. The terms are precise; the discipline of using them properly is itself a learning outcome.
- *“If it’s traditional, it must work.”* — Distinguish empirical claims (testable) from ethical/aesthetic frames (which need separate justification). Encourage students to ask “how would I test this?”.

Differentiation strategies

- **Extension for advanced students:** offer one optional “deeper read” per day — usually a primary source or modern paper.
- **Support for students needing scaffolds:** more pair-work, allow oral responses, provide partial outlines for writing prompts.
- **Multilingual classrooms:** Sanskrit terms are universal; vernacular equivalents (Hindi, Tamil, Telugu, Bengali, Marathi etc.) should be welcomed.

Safety

- Any activity involving food, plants, fire, or outdoor work needs explicit safety briefing.
- See the relevant day file for activity-specific safety notes.

Honest framing

- Where modern science clearly extends or revises traditional claims, say so.
- Where the tradition’s framing is best understood as ethical/aesthetic, present it as such (not as competing science).
- Where attribution is debated (e.g., the “Vedic” status of Tirthaji’s mathematics), present the debate honestly.

End-of-module reflection (for teachers)

- Which concepts landed well? Which didn’t?
- Which students grew most? Which need a different approach next module?
- Add notes to `teacher-notes.md` for the next teacher who uses this pack.

Sources

Sources — Moon Phases and Tithi (Senior)

Primary and secondary sources

- Sūrya-siddhānta (c. 5th–10th c. CE), chapter 2 on lunar motion
- Varāhamihira, *Pañcasiddhāntikā* (6th c. CE)
- NCERT Physics Class XI, ch. 8 (Gravitation)
- NCERT Geography Class VI, ch. 1 (The Earth in the Solar System)
- NASA Lunar Phase educational pages

A note on citation discipline

Per the curriculum’s style guide, citations must be verifiable. If a verse number, page, or attribution cannot be confirmed, the source is paraphrased without invented attribution. Where direct quotes appear in Senior-band day files, the source is identified in IAST + edition.

For deeper reading

- The shared glossary at [curriculum/_shared/glossary.md](#) contains the IAST + meaning for every Sanskrit term used.
- Other modules in this curriculum that connect: see the module README's "Prerequisites" section.

Day 1 — The Moon and Its Shapes

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: observe and name the four basic phases (new, first quarter, full, last quarter).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *candra* (IAST: candra; lit. "Moon")

Primary source (today's reading)

From *Sūrya-siddhānta* (c. 5th–10th c. CE) — see [sources.md](#).

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to "*What's one thing you remember about yesterday's lesson?*"
- Hook: introduce a phenomenon or question that motivates today's concept.

Core (20 min)

1. **Geometric model.** Use a torch (Sun) + tennis ball (Moon) + a student's head (Earth). Walk the Moon around the Earth. "*Which part of the Moon does the Earth-observer see lit at each position?*" Demonstrate the four primary phases.
2. **Define synodic month.** ~29.53 days from one *amāvasyā* to the next. *Synodic* = relative to the Sun (as seen from Earth). Distinct from sidereal month (~27.32 days) — relative to fixed stars.
3. **Source.** *Sūrya-siddhānta*, ch. 2, vv. 1–10 — defines lunar motion with mean and true longitudes. Read 2 verses aloud in transliteration.

4. **Four phases:**

- *amāvasyā* — new (Sun side)
- First quarter — half lit, waxing
- *pūrṇimā* — full (anti-Sun side)
- Last quarter — half lit, waning

5. **Class observation assignment.** Each student notes the moon’s shape this evening (or tomorrow morning) and writes one sentence describing it. We’ll plot data tomorrow.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces The Moon and Its Shapes.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 2 — Two Fortnights — Pakṣa

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: distinguish śukla pakṣa from kṛṣṇa pakṣa and name today’s.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *pakṣa* (IAST: pakṣa; lit. “fortnight (half-month)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define pakṣa.** A *pakṣa* is a half-month: 14–16 days. Two per synodic month.
 - *śukla pakṣa* (waxing fortnight) — *amāvasyā* → *pūrṇimā* — moon growing.
 - *kṛṣṇa pakṣa* (waning fortnight) — *pūrṇimā* → *amāvasyā* — moon shrinking.
2. **Etymology.** *śukla* = bright; *kṛṣṇa* = dark. *Pakṣa* = side / wing — same root as *pakṣapāta* (taking sides).
3. **Source.** Sūrya-siddhānta and the *Pañca-siddhāntikā* of Varāhamihira both organise lunar tables by *pakṣa*.
4. **Identify today’s pakṣa.** Pull up the current *pañcāṅga* (web or printed). “*Which pakṣa are we in right now? Which tithi?*” Students record the date AND the *pakṣa* name in their workbook.
5. **Cultural note.** Many traditional Indian festivals are named by *tithi* within *pakṣa* (e.g. *Karka Saṅkrānti tritīyā*, *Karwa Chauth* = *Kṛṣṇa caturthī* of Kārtika). *Pakṣa* labelling is more precise than “the third of Kārtika.”

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Two Fortnights — *Pakṣa*.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 3 — Fifteen Tithis per Pakṣa

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: recite the 15 tithi names (*pratipadā*, *dvitīyā*, ...) and explain why a tithi isn't exactly a day.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See *lesson-plan.md* for any day-specific materials)

Vocabulary introduced today

- *śukla pakṣa* (IAST: *śukla pakṣa*; lit. “waxing fortnight (new → full)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define tithi precisely.** A *tithi* is the time taken for the Moon to gain 12° of longitudinal separation from the Sun. 30 tithis per synodic month → 15 per pakṣa. A tithi is therefore a unit of angular progress, NOT a unit of time.
2. **Why a tithi ≠ a day.** Lunar speed varies (elliptical orbit, ~3.4°/hour to ~5.5°/hour). A tithi can last 19 to 26 hours. Two consecutive tithis can both touch the same sunrise (*kṣaya tithi*) or one tithi can span two sunrises (*adhika tithi*).
3. **Name the 15 tithis.** *pratipadā*, *dvitīyā*, *ṭṛtīyā*, *caturthī*, *pañcamī*, *ṣaṣṭhī*, *saptamī*, *aṣṭamī*, *navamī*, *daśamī*, *ekādaśī*, *dvādaśī*, *trayodaśī*, *caturdaśī*, *pūrṇimā/amāvasyā*. The 15th is *pūrṇimā* in *śukla pakṣa*, *amāvasyā* in *kṛṣṇa pakṣa*.
4. **Source.** *Sūrya-siddhānta*, ch. 2 (lunar motion); also *Vasiṣṭha-siddhānta* tradition.

5. **Computation exercise.** Given today’s Moon-Sun elongation in degrees (provide one), students compute today’s tithi number. Then compare with the published pañcāṅga.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Fifteen Tithis per Pakṣa.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 4 — Building a Tithi Calendar

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: construct a tithi calendar for the current month using observation and computation.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *kṛṣṇa pakṣa* (IAST: kṛṣṇa pakṣa; lit. “waning fortnight (full → new)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Pull up a current pañcāṅga** (online or printed). Display the next 14 days with tithi, pakṣa, day-of-week.
2. **Manual construction.** Without looking at the pañcāṅga, students attempt to predict the tithi 7 days from today using:
 - Today’s tithi (e.g. *śukla pañcamī*).
 - Approximate rule: ~1 tithi per 24h.
 - Note: actual tithi may be off by ± 1 due to varying lunar speed.
3. **Source.** Modern Drik pañcāṅga uses Sūrya-siddhānta-derived corrections plus modern astronomical constants (NASA JPL ephemerides). Show the students that traditional and modern computation agree to within minutes.
4. **The intercalation challenge.** Adhika māsa (added month) occurs roughly every 32 months when no Sun-saṁkrānti happens within a lunar month. The 2026–2030 adhika māsa calendar is published; pull it up. “*Why is this needed? What would happen without it?*”
5. **Project artefact.** Each student produces a one-month tithi calendar for the CURRENT month, side-by-side with the Gregorian dates. This will be assessed as part of the summative project.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Building a Tithi Calendar.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 5 — Lunar Months and Festivals

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: map the 12 lunar months (caitra, vaiśākha, ...) to major festivals.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *tithi* (IAST: tithi; lit. “lunar day (1/30 of synodic month)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Twelve lunar months.** *Caitra, Vaiśākha, Jyaiṣṭha, Āṣāḍha, Śrāvaṇa, Bhādrapada, Āśvina, Kārtika, Mārgaśīrṣa (Agrahāyaṇa), Pauṣa, Māgha, Phālguna.* Each begins on a new moon (*pūrṇimānta* tradition — North India) OR the day after the previous full moon (*amānta* tradition — South India). Both reference the same lunar month with a 15-day phase difference.
2. **Source.** Varāhamihira’s *Bṛhat Saṃhitā* discusses both reckonings; the Sūrya-siddhānta uses *amānta*.
3. **Festivals by tithi.** Examples:
 - *Diwali* — *Kṛṣṇa amāvasyā* of Kārtika
 - *Holi* — *Pūrṇimā* of Phālguna (followed by *Vasantotsava*)
 - *Janmāṣṭamī* — *Kṛṣṇa aṣṭamī* of Śrāvaṇa
 - *Mahā-Śivarātri* — *Kṛṣṇa caturdaśī* of Māgha

4. **Class mapping.** Hand out cards with festival names. Students place each on a 12-month wheel, noting tithi + pakṣa. Compare two regions' festival calendars.
5. **Honest history.** Not all festivals are pan-Indian. Some are regional (Onam — Malayalam solar calendar; Pongal — Tamil solar calendar). Distinguish lunisolar tradition from purely solar reckonings.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Lunar Months and Festivals.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 6 — Panchanga — Five Limbs

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify the five components: tithi, vāra, nakṣatra, yoga, karaṇa.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *amāvasyā* (IAST: amāvasyā; lit. “new moon”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define pañcāṅga.** Five (*pañca*) limbs (*aṅga*) of the Indian calendar:
 - *tithi* — lunar day (Module focus so far)
 - *vāra* — weekday (7-day cycle, *ravivāra–śanivāra*)
 - *nakṣatra* — lunar mansion (27 divisions of ecliptic)
 - *yoga* — Sun-Moon longitudinal sum (27 named yogas)
 - *karaṇa* — half-tithi (11 named karaṇas)
2. **Source.** Sūrya-siddhānta, *Pañcāṅga-prakaraṇa* sections.
3. **Read a published pañcāṅga together.** Show today’s row. “*Identify each of the five aṅgas.*” Note that the pañcāṅga also lists *muhūrta* (auspicious hours), *rāhu-kāla*, *yamaṅga* — derivative computations.
4. **Modern parallel.** Astronomical observational data (NASA / IMCCE) provide the *raw inputs* (Sun-Moon longitudes); the pañcāṅga *interprets* these through the 5-limb framework. The astronomy is universal; the framework is Indian.
5. **Critical-thinking discussion.** “*Pañcāṅga has been used for ritual timing for ~2000 years. What does that tell us about (a) the durability of the framework and (b) the limits of inferring causation from co-occurrence?*”

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Panchanga — Five Limbs.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 7 — Why Eclipses Happen

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: explain solar/lunar eclipses using orbital plane geometry (5° tilt of moon's orbit).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *pūrṇimā* (IAST: *pūrṇimā*; lit. “full moon”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Geometric setup.** Solar eclipse — Moon between Sun and Earth (occurs at *amāvasyā*). Lunar eclipse — Earth between Sun and Moon (occurs at *pūrṇimā*).
2. **Why not every new/full moon?** The Moon’s orbital plane is tilted ~5.14° from the ecliptic. Eclipses happen only when the Moon is near a *node* (intersection of orbital planes) during new or full moon. This is the *Rāhu/Ketu* node geometry of classical Indian astronomy.
3. **Source.** Āryabhaṭīya, *Gola-pāda* — gives a heliocentric-leaning model and correctly predicts eclipses 1000+ years before modern astronomy.
4. **Show a recent eclipse path map** (any annular or total eclipse from the last 5 years). “*Why does the path strike only a thin ribbon, not the whole hemisphere?*” (Geometric umbra/penumbra.)

5. **Safety rule.** Solar eclipse — NEVER look at the Sun directly. Use projection or eclipse-rated filters only. Lunar eclipse is always safe. State this rule on the board and have students repeat.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Why Eclipses Happen.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 8 — Modern Lunar Science

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: phases as sun’s illumination of moon; mention libration, tides.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *pañcāṅga* (IAST: *pañcāṅga*; lit. “five-limbed Indian calendar (tithi, vāra, nakṣatra, yoga, karaṇa)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Phases re-explained.** Lunar phases = sunlight reflection from a 1737-km-radius sphere as the Earth-Moon-Sun angle changes. Synodic period of ~29.53 days. There is no “dark side” — only a *far side*, which receives the same sunlight, just not visible from Earth.
2. **Libration.** Earth-observers see ~59% of the Moon’s surface over time due to orbital eccentricity and axial tilt. Demo with rotating ball.
3. **Tidal lock.** Same face always visible to Earth — rotation period = orbital period = ~27.32 days (sidereal). This is the result of tidal forces over billions of years.
4. **Modern lunar exploration.** Mention briefly: Apollo missions (1969–72), Chandrayaan-1 (2008) discovered water-ice signatures, Chandrayaan-3 (2023) soft-landing near south pole, Artemis program (2024+).
5. **Bridge back to tradition.** Sūrya-siddhānta computes Moon’s distance and diameter approximately (Earth-Moon ~38000 yojanas $\approx$ 380000 km — close to modern 384400 km). The empirical precision is striking; the cosmological framework around it is what differs.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Modern Lunar Science.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.

- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 9 — Cross-cultural Calendars

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compare with hijri (islamic), chinese, and hebrew lunar calendars.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *candra* (IAST: candra; lit. “Moon”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Indian lunisolar.** Lunar months (~29.5d) + adhika māsa intercalation to track the solar year. Months named for the *nakṣatra* near the full moon.
2. **Islamic Hijri.** Strictly lunar — 12 lunar months, no intercalation. ~354 days/year, drifts ~11 days/year relative to solar calendar. Reasons: religious lunar observation, no agricultural anchoring.
3. **Chinese.** Lunisolar like Indian, with intercalation. 60-year cycle combining 12 animals × 10 stems. Confucian calendar reform 104 BCE refined it.
4. **Hebrew.** Lunisolar with intercalation (Adar I/II in leap years). 19-year Metonic cycle.
5. **Roman / Gregorian.** Purely solar (since Julian 45 BCE, refined 1582 CE). Months disconnected from lunar cycle. Most universal calendar today.
6. **Source comparison.** Show a 2026 calendar in 4 traditions side-by-side. “*Which dates align? Which don’t? Why?*”

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Cross-cultural Calendars.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 10 — Synthesis and Quiz

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: final quiz and 10-day moon-diary share.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *pakṣa* (IAST: pakṣa; lit. “fortnight (half-month)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Vocabulary lightning round.** Students take turns defining: *tithi*, *pakṣa*, *śukla*, *kṛṣṇa*, *pūrṇimā*, *amāvasyā*, *pañcāṅga*, *nakṣatra*, *adhika māsa*. Catch errors collectively.
2. **Moon-diary review.** Each student shares one observation from their 10-day moon diary (Days 1–10 of homework). Identify discrepancies between prediction and observation.
3. **Source recall.** “Name one classical source and one modern source we cited for this module.” (Sūrya-siddhānta + NCERT / NASA.)
4. **Critical-thinking close.** “The Indian lunisolar calendar is ~2000+ years old and still in active use. The Gregorian calendar is ~450 years old. Why might multiple calendars coexist in modern India?” (Cultural specificity + scientific universality.)
5. **Summative quiz** (45 min, written) — see `quizzes/summative.md`.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Synthesis and Quiz.

Wrap-up (5 min)

- One-sentence exit ticket: “What’s one thing you learned today?”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Formative Quiz — Moon Phases and Tithi (Senior)

When: mid-module (after Day 5). **Duration:** 25 minutes. **Format:** mix of short answer, multiple choice, and reflection.

Questions

1. Name the three / four key terms introduced in this module so far. Give the IAST and one-line meaning of each.

2. State the central idea of Moon Phases and Tithi in one sentence.
3. (Multiple choice) Which day in this module focused on ‘Fifteen Tithis per Pakṣa’? (a) Day 1 (b) Day 2 (c) Day 3 (d) Day 4
4. True / False — and explain in one sentence: *Moon Phases and Tithi is purely a religious concept and has no scientific framing.*
5. Connect Moon Phases and Tithi to **one** other module in this curriculum. Which module, and how?
6. List **two** sources cited so far in this module.
7. (Short answer) Why does the module distinguish ‘candra’ from related concepts? Give one example.
8. (Short answer) In your own words, explain Day 3’s concept.
9. Identify **one** common misconception about Moon Phases and Tithi and rebut it in one sentence.
10. Write **one** question you have about this module so far — something you want clarified before the summative.

How this is used

This quiz is **formative** — it’s a check-in to see what’s landing, not a grade. Teacher reviews answers to adjust pacing for Days 6–10. Students may swap and self-mark.

Summative Quiz — Moon Phases and Tithi (Senior)

When: end of module (after Day 10). **Duration:** 45 minutes. **Total points:** 30.

Questions

Part A — Vocabulary (~5 points) A1. Define *candra* in your own words. A2. Define *pakṣa* in your own words. A3. Define *śukla pakṣa* in your own words. A4. Define *kṛṣṇa pakṣa* in your own words. A5. Define *tithi* in your own words.

Part B — Concepts (~10 points) B1. Explain the central idea of Moon Phases and Tithi in your own words (3–4 sentences). B2. Describe the method or framework students used in Day 4 of this module. B3. Identify ONE source cited in this module and explain why it matters. B4. Compare Moon Phases and Tithi with a concept from another module (your choice). What’s similar, what’s different? B5. Identify ONE common misconception about this module and rebut it in 2–3 sentences.

Part C — Application (~5 points) C1. Give an example of Moon Phases and Tithi from your own daily life. C2. Describe one way this module connects to a current real-world issue (or scientific question).

Part D — Reflection (~5 points; ungraded for content) D1. What surprised you most in this module? D2. What’s one question you still have?

Part E — Primary source (~5 points; Senior only) E1. Quote or paraphrase one passage from a primary source cited in this module. Explain it in your own words.

Marking guidance

- Vocabulary: give credit for correct meaning, even if exact wording differs.
- Concepts: look for evidence of understanding, not memorisation.
- Application: any reasonable example accepted.
- Reflection: ungraded for content; awarded for thoughtfulness.

Pass mark

A score of 18/30 indicates the student has met the module's learning objectives.

Homework

Homework — Moon Phases and Tithi (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Moon and Its Shapes** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Two Fortnights — Pakṣa** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Fifteen Tithis per Pakṣa** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Building a Tithi Calendar** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Lunar Months and Festivals** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Panchanga — Five Limbs** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Why Eclipses Happen** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Modern Lunar Science** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Cross-cultural Calendars** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Synthesis and Quiz** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

Writing Prompts

Writing Prompts — Moon Phases and Tithi (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Moon Phases and Tithi to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Moon Phases and Tithi with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Moon Phases and Tithi? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are doṣas a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Activity 1 — 10-Day Moon Observation Diary

Module: Moon Phases and Tithi · **Band:** Senior · **Duration:** 30 min

Purpose

Reinforce the day-by-day concepts of Moon Phases and Tithi through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

notebook, pencils, ~15 minutes per evening

What students do

Each evening for 10 days, students sketch the moon shape they see, note the time, the tithi name, and one feeling/observation. On Day 10 they compare sketches with the actual calendar.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (20 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Project Brief

Project Brief — Class Tithi Calendar for the Current Month

Module: Moon Phases and Tithi · **Band:** Senior

What you will do

Each student picks 3 dates in the current calendar month. They compute (or look up) the tithi, identify the pakṣa, and produce a visual class calendar with all entries combined.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a poster / model / digital deliverable).
- **Written reflection** (500 words (Senior)): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (5-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 3 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

Rubric

Assessment Rubric — Moon Phases and Tithi (Senior)

This rubric applies to the project (see `project-brief.md`) and may also inform the summative quiz.

Criteria

Accuracy of concepts — uses Sanskrit terms correctly, gets the central idea right

- **Exceeding (4)** — Exceptional; could teach this to another student.

- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Source discipline — cites sources, distinguishes traditional framing from empirical claim

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Application — connects the module to other modules and to real-world examples

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Presentation — clear, organised, evidence-supported

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Reflection — honest, thoughtful, identifies what was learned and what remains uncertain

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

How the teacher uses this rubric

1. After the project share, score each student on each criterion.
2. Convert to a band (4 = Exceeding, 3 = Meeting, etc.) for the report card.
3. Share at least one strength and one area for growth with each student.

Self-assessment (Senior students)

Senior students complete a parallel self-assessment using the same rubric before the teacher's assessment. The teacher and student then meet to compare.